

SALIENT FAUSTA MAKIA NKONGHO ENOH

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EDUCATION

Widener University

Chester, PA

B.S. Robotics, Mechatronics & Automation Engineering • GPA: 3.65/4.0

Expected May 2027

Minor: Computer Science

Honors: Widener Presidential Scholarship, PECO Engineering Scholarship, Dean's list, Honors in Engineering

Academic Topics: Robotics, Mechatronics, Autonomous Navigation, Embedded Systems, Camera Vision, Operating Systems, Sensor Fusion, Control Systems, MATLAB, C++, Python, AutoCAD, Linear Algebra

TECHNICAL SKILLS

Programming & Analysis: Python, C++, Embedded C, MATLAB, Linux (Ubuntu), Bash, Git

Data & Sensor Analysis: Signal filtering, noise reduction, statistical analysis, time-series data processing, sensor data validation, log analysis

Sensors & Instrumentation: LiDAR, IMU, GPS, ultrasonic, thermal sensors, cameras, encoders, pressure/depth sensors

Robotics Systems: ROS1/ROS2, sensor fusion (Kalman filtering), state estimation, PID control, real-time systems

Hardware: Arduino, Raspberry Pi, STM32, ESP32, NVIDIA Jetson, soldering, circuit debugging

Testing & Validation: Hardware-in-the-loop (HIL), system debugging, simulation (MATLAB/Simulink, RViz), field testing, performance evaluation

Mechanical & Fabrication: SolidWorks, AutoCAD, FEA, rapid prototyping, enclosure design, 3D printing, mechanical assembly

Certification: Universal Robot e-Series Core track - January 2026

ENGINEERING EXPERIENCE

WIDENER UNIVERSITY - Robotics, AI and Automation Research Laboratory

Chester, PA | 2024–Present

Semi-Autonomous Electric Wheelchair (Robotic Vehicle Platform)

May–September 2025

- Built a ROS2-based real-time multi-sensor data processing software (Python) on NVIDIA Jetson for autonomous navigation
- Designed a multi-modal data pipeline integrating LiDAR, camera, IMU, and ultrasonic inputs for environmental perception
- Processed camera and LiDAR into point clouds and filtered sensor streams for obstacle detection and motion planning
- Developed perception across 8+ sensors filtering through hundreds of sensor data every cycle and making accurate decisions
- Developed sensor fusion algorithms distance-based decision logic and obstacle detection systems for real-time navigation and path planning
- Designed and built custom collision resistant mechanical mounts using AutoCAD, FEA, 3D printing and laser cutting
- Conducted simulation (MATLAB, Simulink, Rviz), field testing and iterative debugging, improving system perception accuracy by ~75%
- Awarded Best Research Presentation for delivering a fully functional autonomous mobility system.

Autonomous Underwater Vehicle (RoboSub 2025)

May–September 2025

- Developed embedded software (C++/Python, Linux) for real-time sensor acquisition and control
- Integrated camera, IMU, and depth sensors to estimate depth and orientation in a challenging underwater environment
- Built vision-based perception pipelines (OpenCV) for underwater object detection under low-visibility conditions
- Designed and tuned PID control systems using sensor feedback, stabilizing depth and heading
- Identified and mitigated sensor drift and environmental distortions in underwater measurements
- Designed and fabricated a pressure-resistant enclosure, ensuring reliable sensor operation in harsh conditions
- Fabricated custom components using AutoCAD, 3D printing and laser cutting ensuring proper fit of electrical components
- Collaborated across multidisciplinary teams in an international robotics competition

Autonomous Firefighting Robot Vehicle

January – May 2025

- Built an autonomous robot with real-time sensor data processing and decision-making capabilities (Arduino, Raspberry Pi)
- Developed embedded software (C++/Python) for continuous sensor integration across thermal and ultrasonic sensors
- Achieved sensor-based navigation where the robot dynamically plans its trajectory and moves towards the fire source using a thermal sensor
- Implemented PID control based on encoder feedback for smooth and precise navigation

AI & Machine Learning Researcher

May - September 2024

- Designed and deployed a multimodal AI Chatbot system integrating computer vision and speech recognition,
- Developed vision-based facial recognition and tracking pipelines using camera input to recognize specific users
- Implemented speech-to-text and text-to-speech systems for real-time human interaction
- Integrated LLM models and APIs, engineered in Python on Linux, optimized on user feedback

Robotics, Automation and Controls Research Laboratory Assistant

January – May 2024

- Assisted with mechanical assembly and programming of a humanoid robot for human interaction and live demonstrations
- Supported sensor installation, actuator integration, and firmware debugging during robotics system testing

LEADERSHIP EXPERIENCE (2024 – Present)

Founder & President, Widener AI Club: Founded and lead a 40+ member focused on robotics, AI, and engineering workshops

